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Problem Set 1

Advanced Macroeconomics

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Get Started

Open Octave or Matlab, create a new file getstart.m and write a code for the following operations:

1. Matrices
   (a) Enter a matrix with numerical values:

$$A = \begin{pmatrix} 16 & 3 & 2 & 13 \\ 5 & 10 & 11 & 8 \\ 9 & 6 & 7 & 12 \\ 4 & 15 & 14 & 1 \end{pmatrix}$$

   (b) Calculate the sum per column.
   (c) Transpose the matrix.
   (d) Calculate the sum per row of the matrix.
   Note: do same as for column sums but with transposed matrix to get row sums and transpose the result itself to display rows.
   (e) Display the diagonal of the matrix.
   (f) Calculate the sum of the diagonal elements.
   (g) Calculate the sum of all elements in the 4th column.
   Note: extract single elements by $A(row, column)$.
   (h) Use the colon operator to display
   Note: colon operator is the ":".
   - the numbers from 1 to 10.
   - numbers backwards from 100 to 50 in steps of 7.
   - elements 1 to 4 of the 4th column of matrix A.
   - elements 1 to 4 of the last column of matrix A.

2. Expressions

(a) Define a variable with the number of students in the course.

(b) Define a variable $aa = 2 + 2$ and use it for the following calculations:

i. $bb = aa - 1$

ii. $cc = aa \times bb$

iii. $dd = cc/aa$

iv. $ee = aa^3$

v. $ff = (aa + bb)cc$

vi. $gg = (1 + \sqrt{5})/2$

vii. $hh = \ln(gg)$
Note: natural logarithm; log of x with base y would be $log(x)/log(y)$.

viii. $ii = \exp(hh)$

3. Working with matrices

(a) Matrix generation

- Generate a matrix Z with 2 rows and 4 columns filled with zeros.
- Generate a matrix F with 3 rows and 3 columns filled with fives.
- Generate a matrix N with 10 random whole numbers between 0 and 9.
  Note: $rand(1, 10)$ creates a vector with 10 random numbers between (incl.) 0 and (excl.) 1; $10 * ...$ scales the interval to [0, 10), $fix()$ cuts the decimals.
- Generate a 4×4 matrix R with random numbers given the normal distribution.
  Note: $rand()$ would be equal distribution and generate value between (incl.) 0 and (excl.) 1.

(b) Element-wise operation

- Do an element-wise exponentiation of N with exponent 2.
- Do an element-wise multiplication of A with A.
  Note: $N.*N$ would be same as above; both matrices must have same dimension because the single elements are multiplied.

(c) Create a concatenated matrix B of fourfold dimension of matrix A.

(d) Create a matrix $X = A$ and delete the second column.

(e) Linear algebra

- Add matrix A' to matrix A.
- Multiply matrix A' with matrix A.
  Note: matrix multiplication

$$A \times B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix},$$

where

$$
\begin{aligned}
c_{11} &= a_{11}b_{11} + a_{12}b_{21}, \\
c_{21} &= a_{21}b_{11} + a_{22}b_{21}, \\
c_{12} &= a_{11}b_{12} + a_{12}b_{22}, \\
c_{22} &= a_{21}b_{12} + a_{22}b_{22}.
\end{aligned}
$$

- Calculate the inverse of matrix R.
  Note: $R^{-1} * R = I = [10; 01]$.
  The inverse matrix is a square matrix that multiplied with the original matrix yields the identity matrix. Not every quadratic matrix is invertible. (Invertible matrices are regular matrices. They must build a bijective linear mapping.)
- Multiply the inverse by the original matrix.
  Note: the result should be the identity matrix.
- Define a matrix v with 4 rows and 1 column (column vector), consisting of ones.
- Multiply matrix A by v.
  Note: gives weighted sums of column in A (here weight = 1).

4. Arrays

   (a) Build a 10×3 table with different numerical values. Create first a column vector n with values from 0 to 9. Then, apply a formula using the respective element of n (n^2 and 2^n, e. g.) to fill the other two columns with numerical values.

   (b) Build a table that contains values of a variable x in the first column and its logarithm base 10 in the second column.

   (c) Create a matrix

$$
D = \begin{pmatrix} 72 & 134 & 3.2 \\ 81 & 201 & 3.5 \\ 69 & 156 & 7.1 \\ 82 & 148 & 2.4 \\ 75 & 170 & 1.2 \end{pmatrix}.
$$

   Calculate mean μ and standard deviation σ of every column.
   Note: mean of every column; $mean(D, 2)$ = mean of every row; or $std(D, 0, 2)$.

   (d) Use scalar expansion to create a new matrix B by subtracting 8.5 from each element of A.
   Note: expansion of scalar $s = 8.5$ by elementwise operation.

   (e) Set the elements B_{12} to B_{23} to zero.

(f) Define a vector

$$x = [2.1, 1.7, 1.6, 1.5, \text{NaN}, 1.9, 1.8, 1.5, 5.1, 1.8, 1.4, 2.2, 1.6, 1.8],$$

and display all finite values of the vector.
Note: logical subscripting means true or false.

(g) Keep only the elements of x that are finite numbers.
Note: try isfinite(x); this gives you instead of the values in x the binary numbers for true and false.

(h) Keep only the elements of x that absolutely deviate less than 3 standard deviations from the mean (remove outliers).

(i) Redefine the matrix B as $B = A$. Overwrite all numbers in B that are not prime with zeros.

(j) Define a row vector k with the position of all elements in A that are prime. Display those elements of A. After that, set them to NaN.
Note: k is row vector; $A(k)$ gives number of elements in A that are prime.

5. Controlling Input and Output

- Create a vector $x = [4/3, 1.2345 \exp(-6)]$ and display it using different formats.
- Create a magic 10×10 matrix A and suppress the output.
  Note: suppressing output means the use of ; after the command; magic matrix = sum of every row, column and of both diagonals is the same = magic constant.
- Enter a long statement with line brake.

6. Graphics

- Create a plot of $y = \sin(x)$. Add axis labels and a plot title.
- Create a plot with several functions $\sin(x)$, $\sin(x - 0.25)$, and $\sin(x - 0.5)$ and add a legend.
- Open a separate figure window for your sinus function.
- Clear an existing figure object for a new plot.
- Create a plot with several subplots. You can choose $\sin(x)$, $\cos(x)$, $\ln(x)$, and $\exp(x)$, for example.
- Discover and try other features of plot settings.

7. Programming

- Define $a = 1$ and $b = 2$. Using the if-statement, check if a is greater, equal, or smaller than b. Display the result of the check in the command window.
- Write a for-loop to display values of n from 1 to 10 in the command window.

- Create a structure containing name and year of birth of famous economists.
- Write a separate function and use it to quickly calculate x^2.
- Create a so-called anonymous function for x^2.
  Note: this is a quicker alternative to functions with a separate file - on the other hand, this is a local function that can only be used within this code file; separate functions are global and can be called by other scripts.

8. Environment Identification

   - Check whether you are using Octave or Matlab.
     Note: $exist()$ checks if a certain variable exists; this specific built-in variable can only exist in the GNU Octave environment, so if it's not 0, then we identify our environment as Octave; if this is not the case, we define our environment as Matlab. A built-in variable in Matlab or Octave is a variable that is predefined by the system and exists as part of the core environment. These variables are not created by the user, and they often provide system-level information or functionality.
   - Note: This is useful since in some cases, commands differ between Octave and Matlab. In such cases, you can use an if condition where first the environment is checked, and after that you can run the respective command.